

Probability

Random Variable :-

A Random variable may be defined as a real valued function which takes real values determined by the outcomes of the random experiment.

Note:-

- (1) A random variable is denoted by X .
- (2) $P(X=a) \rightarrow$ Probability of the event X at a value 'a'.
- (3) $P(a < X < b) \rightarrow$ Probability of the event X at any value in the interval

Random Variable

Discrete
Random Variable



If a random variable X assumes only a finite set of values it is known as D.R.V.

E.g:- Marks obtained by students in a test.

Continuous
Random Variable



If a random variable X assumes an infinite no. of uncountable values it is known as C.R.V.

E.g:- Height or weight of students in a class.

Probability Distribution of a Discrete Random Variable

Let X be a discrete random variable. If the probability for $X = x_i$ is p_i then

$$P(X = x_i) = p_i = p(x_i) \text{ for } i = 1, 2, \dots$$

where p_i

(i) $p(x_i) \geq 0$ for all values of i

(ii) $\sum p(x_i) = 1$

The function p_i is called probability function or probability mass fⁿ (p.m.f) of the random variable X .

The set of all possible ordered pairs $\{x_i, p(x_i)\}$ is called probability distribution of the random variable X .

x	x_1	x_2	x_3	- - -	x_n
$p(x)$	p_1	p_2	p_3	- - -	p_n

$$\text{Mean} = \sum p_i x_i = \mu$$

$$\text{Variance} = \sum p_i x_i^2 - \mu^2$$

(σ^2)

$$\text{Standard Deviation}(\sigma) = \sqrt{\text{Variance}}$$

(3)

Q:- A random variable X has the following probability function

$x :$	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) Find the value of k

(ii) Find $P(X < 6)$

Solⁿ:- (i) $\sum_{i=0}^7 p(x_i) = 1$

$$\Rightarrow p(x_1) + p(x_2) + p(x_3) + p(x_4) + p(x_5) + p(x_6) + p(x_7) = 1$$

$$\Rightarrow 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 0$$

$$\Rightarrow 10k^2 + 10k - k - 1 = 0$$

$$\Rightarrow 10k(k+1) - 1(k+1) = 0$$

$$\Rightarrow (k+1)(10k-1) = 0$$

$$\Rightarrow k = -1, \frac{1}{10}$$

[negative value not possible]

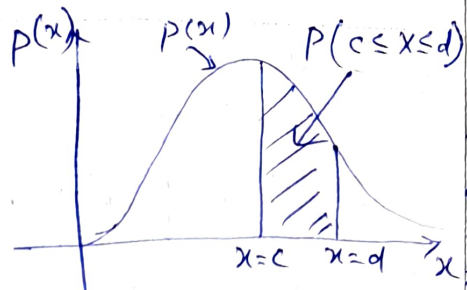
$$\therefore k = \frac{1}{10}$$

$$\begin{aligned} \text{(ii)} \quad P(X < 6) &= P(X=0) + P(X=1) + P(X=2) \\ &\quad + P(X=3) + P(X=4) + P(X=5) \\ &= 0 + \frac{1}{10}k + 2k + 2k + 3k + k^2 \\ &= k^2 + 8k = \left(\frac{1}{10}\right)^2 + \frac{8}{10} = \frac{81}{100} \end{aligned}$$

Probability Distribution of a Continuous Random Variable

Let X be a continuous random variable taking values on the interval $[a, b]$.

A function $p(x)$ is said to be the probability density function of the continuous random variable X if it satisfies the following properties:



- (i) $p(x) \geq 0$ for all x in the interval $[a, b]$.
- (ii) For two distinct numbers c & d in the interval $[a, b]$,
 $P(c \leq X \leq d)$ = Area under the probability curve between the ordinates at $x=c$ & $x=d$
- (iii) Total area under the probability curve is 1
 i.e., $P(a \leq X \leq b) = 1$.

Note :-

$$(1) P(c \leq X \leq d) = \int_c^d p(x) dx$$

- (2) For a continuous random variable the probability at a point is always zero
 i.e., $P(X=c) = 0$, for all single point values of c .

$$(3) P(c \leq X \leq d) = P(c < X \leq d) \\ = P(c \leq X < d) = P(c < X < d)$$

But this is not true if X is a discrete random variable.

Distribution Function or Cumulative Probability Function

(1) If X is a discrete random variable with probability function $P(x)$ then, the distributive function $F(x)$ is defined as

$$F(x) = P(X \leq x)$$

(2) If X is a continuous random variable with p.d.f $p(x)$, then the distribution function $F(x)$ is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x p(x) dx$$

Q:- A die is tossed twice. Getting 'an odd number' is termed as a success. Find the probability distribution of the no. of success.

Solⁿ:- Since 'getting an odd number' is success

$$\therefore \text{Probability of success (S)} = \frac{3}{6} = \frac{1}{2}$$

$$\text{Probability of failure (F)} = 1 - \frac{1}{2} = \frac{1}{2}$$

If X denotes the no. of successes in two throws of a die, then

$$P(X=0) = \text{Probability of no success} \\ = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(X=1) = \text{Probability of one success} \\ = P(S)P(F) + P(F)P(S) \\ = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

(6)

$$P(X=2) = \text{Probability of two successes}$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Probability distribution of X is given by

x	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Mathematical Expectation

If X is a random variable which can assume any one of the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$, then the mathematical expectation of X denoted by $E(X)$ is defined as

$$E(X) = p_1 x_1 + p_2 x_2 + \dots + p_n x_n$$

$$= \sum p_i x_i = \mu \text{ (Mean)}$$

where μ = mean of the probability distribution of a variable X

(*) For discrete distribution

$$E(X) = \sum p_i x_i$$

(*) For continuous distribution

$$E(X) = \int_{-\infty}^{\infty} x p(x) dx$$

⑦

Variance of a distribution is given by

$$\sigma^2 = \sum (x_i - \mu)^2 p(x_i) \quad [\text{Discrete distribution}]$$

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx \quad [\text{Continuous distribution}]$$

Standard Deviation = $\sqrt{\text{Variance}}$

r^{th} moment about the mean is defined by

$$\mu_r = \sum (x_i - \mu)^r p(x_i) \quad [\text{Discrete distribution}]$$

$$\mu_r = \int_{-\infty}^{\infty} (x - \mu)^r p(x) dx \quad [\text{Continuous distribution}]$$

Note :-

(1) $E(c) = c$, where c is a constant

(2) $E(cX) = c E(X)$, where c is a constant

(3) $E(aX + b) = a E(X) + b$, where a & b are constants

(4) $E(X + Y) = E(X) + E(Y)$
where X & Y are random variables

(5) $E(XY) = E(X) \cdot E(Y)$

(6) $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$

(7) $E(X_1 X_2 \dots X_n) = E(X_1) E(X_2) \dots E(X_n)$

(8) $E(X^2) = \sum x_i^2 p_i$

$$\text{Variance}(\sigma^2) = E(X^2) - [E(X)]^2$$

Q:- A die is tossed twice. Getting 'a number greater than 4' is considered a success. Find mean & variance of the probability distribution of the number of successes.

Sol:- Probability of success = $\frac{2}{6} = \frac{1}{3}$

Probability of failure = $1 - \frac{1}{3} = \frac{2}{3}$

Let X denote the no. of successes.
Then

$$P(X=0) = \text{Prob. of no success} \\ = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

$$P(X=1) = \text{Prob. of one success \& one failure} \\ = \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{3} = \frac{4}{9}$$

$$P(X=2) = \text{Prob. of two successes} \\ = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

Prob. distribution is given by

X	0	1	2
$P(X)$	$\frac{4}{9}$	$\frac{4}{9}$	$\frac{1}{9}$

$$\text{Mean} = E(X) = \sum x p(x) \\ = 0 \times \frac{4}{9} + 1 \times \frac{4}{9} + 2 \times \frac{1}{9} = \frac{6}{9}$$

$$\text{Variance} = E(X^2) - [E(X)]^2 \\ = \left(0^2 \times \frac{4}{9} + 1^2 \times \frac{4}{9} + 2^2 \times \frac{1}{9} \right) - \left(\frac{6}{9} \right)^2 \\ = \frac{8}{9} - \frac{36}{81} = \frac{4}{9}$$

(9)

Mean deviation from the mean is given by

$$\frac{1}{N} \sum |x_i - \mu| p(x_i) \quad [\text{Discrete distribution}]$$

$$\text{or } \frac{1}{N} \int_{-\infty}^{\infty} |x - \mu| p(x) dx \quad [\text{Continuous distribution}]$$

$$\text{where } N = \sum p(x)$$

Q:- If X is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} kx & , 0 \leq x < 2 \\ 2k & , 2 \leq x < 4 \\ -kx + 6k & , 4 \leq x < 6 \end{cases}$$

Find k and mean value of X .

Sol:- $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (-kx + 6k) dx = 1$$

$$\Rightarrow k \left[\frac{x^2}{2} \right]_0^2 + 2k \left[x \right]_2^4 + \left[-\frac{kx^2}{2} + 6kx \right]_4^6 = 1$$

$$\Rightarrow 2k + 4k + (-18k + 36k + 8k - 24k) = 1$$

$$\Rightarrow 50k - 42k = 1$$

$$\Rightarrow 8k = 1 \Rightarrow k = \frac{1}{8}$$

$$\text{Mean} = E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^2 \frac{1}{8} x^2 dx + \int_2^4 \frac{1}{4} x dx + \int_4^6 \left(-\frac{1}{8} x^2 + \frac{3}{4} x \right) dx$$

$$= 3$$

Moment Generating Function

(1) The moment generating function (m.g.f) of the discrete probability distribution of X about the value $x=a$ is denoted by $M_a(t)$ & is defined as

$$M_a(t) = \sum p_i e^{t(x_i - a)}$$

which is a f^n of parameter t only

$$\begin{aligned} M_a(t) &= \sum p_i e^{t(x_i - a)} \\ &= \sum p_i + t \sum p_i (x_i - a) + \frac{t^2}{2!} \sum p_i (x_i - a)^2 + \dots + \frac{t^r}{r!} \sum p_i (x_i - a)^r + \dots \\ &= 1 + t \mu_1' + \frac{t^2}{2!} \mu_2' + \dots + \frac{t^r}{r!} \mu_r' + \dots \end{aligned}$$

where $\mu_r' = \sum p_i (x_i - a)^r$ is the moment of order r about a .

$$\text{Also } \mu_r' = \left[\frac{d^r}{dt^r} M_a(t) \right]_{t=0}$$

$$\therefore M_a(t) = e^{-at} \sum p_i e^{tx_i}$$

$$\text{or } M_a(t) = e^{-at} M_0(t)$$

(2) If $f(x)$ is the density function of a continuous variate X , then the m.g.f. about $x=a$ is given by

$$M_a(t) = \int_{-\infty}^{\infty} e^{t(x-a)} f(x) dx.$$

Note:

~~(1) $\mu_1' = \frac{d}{dt} M$~~

(1) $M_0(t) = \sum p_i e^{tx_i}$ [Discrete]

or $M_0(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$ [Continuous]

(2) $\mu_1' = \left[\frac{d}{dt} M_0(t) \right]_{t=0} = \text{Mean}$

(3) $\mu_2' = \left[\frac{d^2}{dt^2} M_0(t) \right]_{t=0}$

(4) $\mu_2 = \mu_2' - (\mu_1')^2 = \text{Variance}$

(5) ~~$E(e^{tx})$~~ $E(e^{tx}) = \text{m.g.f. for } x$

Q:- Find the m.g.f. of the exponential distribution

$$f(x) = \frac{1}{c} e^{-x/c}, \quad 0 \leq x < \infty, \quad c > 0$$

Hence find its mean & S.D.

Solⁿ:- The moment generating fⁿ about the origin is

$$M_0(t) = \int_0^{\infty} e^{tx} \cdot \frac{1}{c} e^{-x/c} dx$$

$$= \frac{1}{c} \int_0^{\infty} e^{(t-\frac{1}{c})x} dx$$

$$= \frac{1}{c} \left[\frac{e^{(t-\frac{1}{c})x}}{(t-\frac{1}{c})} \right]_0^{\infty}$$

$$= -\frac{1}{c} \cdot \left(\frac{e}{1-ct} \right) [0-1] \quad [\because |t| < \frac{1}{c}]$$

$$= \frac{1}{1-ct} = (1-ct)^{-1}$$

$$= 1 + ct + c^2 t^2 + c^3 t^3 + \dots$$

$$\therefore \text{Mean} = \mu_1' = \left[\frac{d}{dt} M_0(t) \right]_{t=0}$$

$$= [c + 2c^2 t + 3c^3 t^2 + \dots]_{t=0}$$

$$= c$$

$$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t) \right]_{t=0}$$

$$= [2c^2 + 6c^3 t + \dots]_{t=0}$$

$$= 2c^2$$

$$\text{Variance} = \mu_2 = \mu_2' - (\mu_1')^2$$

$$= 2c^2 - c^2 = c^2$$

$$S.D = \sqrt{\text{Variance}} = c$$